Eric L. Denovellis, Ph.D.

CONTACT INFORMATION

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Google Scholar: Eric Denovellis

RESEARCH INTERESTS

I study how the brain represents past events and alternative possibilities, and how these representations support memory and decision-making. I develop statistical models to measure them: marked point process switching state space models that decode hippocampal replay and classify its dynamics. With collaborators, I adapted these models for real-time closed-loop neurofeedback, a brain-machine interface for hippocampal representations. I build open-source software that makes these methods reproducible across labs.

APPOINTMENTS

University of California, San Francisco, San Francisco, California USA

Research Scientist **September 2019 – Present**
Supervisor: Loren M. Frank, Department of Physiology

- Developed marked point process switching state space models that decode hippocampal replay from clusterless spikes, classify its dynamics, and distinguish among environments
- Adapted these models for real-time decoding with collaborators, enabling closed-loop neurofeedback on hippocampal spatial representations
- Co-led development of Spyglass, an open-source framework for reproducible, shareable analysis; lead the two computational scientists who maintain it
- Collaborate with and mentor trainees on statistical analyses

Publications: *Nature*; *Neuron* (4); *eLife* (3, including first and co-first author); *eNeuro*

Boston University, Boston, Massachusetts USA

Postdoctoral Researcher **September 2016 – August 2019**
Supervisors: Uri T. Eden and Mark A. Kramer, Department of Mathematics and Statistics

- Studied communication between the hippocampus and other brain areas during the reactivation of memory
- Developed hybrid point process state space models for characterizing hippocampal replay dynamics, work continued at UCSF
- Created spectral_connectivity, a package for multitaper spectral estimation and frequency-domain connectivity analysis; GPU support and publication followed at UCSF (2022)

Publications: *J. Open Source Softw.*; IEEE Asilomar Conference (both first author)

Graduate Student **September 2009 – September 2016**

- Identified synchronous oscillatory neural ensembles in prefrontal cortex that support switching between task rules
- Applied generalized additive models and point process methods to characterize neural encoding of cognitive variables

Publications: *Neuron* (co-first author)

Education

Boston University, Boston, Massachusetts USA
Graduate Program for Neuroscience
Ph.D. in Computational Neuroscience, September 2016
Thesis: Task Switching in the Prefrontal Cortex
Advisor: Daniel H. Bullock
Graduate coursework: MA 576 Generalized Linear Models; MA 584 Multivariate Statistical Analysis; MA 585 Time Series and Forecasting; MA 666 Advanced Modeling and Data Analysis in Neuroscience; EC 516 Digital Signal Processing; CN 700 Computational and Mathematical Methods in Neural Modeling

University of California, Santa Barbara, Santa Barbara, California USA
B.S., Mathematics and B.A., Philosophy with High Honors, June 2007
Phi Beta Kappa, Lambda Chapter

Publications

* *Co-first authors*

Peer-reviewed Articles

1. Lee, K.H.*, **Denovellis, E.L.***, Ly, R., Magland, J., Soules, J., Comrie, A.E., Gramling, D.P., Guidera, J.A., Nevers, R., Adenekan, P., et al. (2026). Spyglass is a framework for reproducible and shareable neuroscience research. *eLife*, 14, RP108089. doi:10.7554/eLife.108089.3

2. Comrie, A.E., Monroe, E.J., Kahn, A.E., **Denovellis, E.L.**, Joshi, A., Guidera, J.A., Krausz, T.A., Berke, J.D., Daw, N.D., and Frank, L.M. (2026). Hippocampal representations of alternative possibilities are flexibly generated to meet cognitive demands. *Neuron*. doi:10.1016/j.neuron.2026.08.023

3. Guidera, J.A., Gramling, D.P., Comrie, A.E., Joshi, A., Tseng, S.Y., **Denovellis, E.L.**, Smyth, C.N., Lee, K.H., Zhou, J., Thompson, P., et al. (2026). Regional specialization in prefrontal cortex manifests in the reliability of task progression codes. *Neuron*, 114(6), 1152–1161.e8. doi:10.1016/j.neuron.2026.01.023

4. Chu, J.P., Coulter, M.E., **Denovellis, E.L.**, Nguyen, T.T.K., Liu, D.F., Deng, X., Eden, U.T., Kemere, C.T., and Frank, L.M. (2025). RealtimeDecoder: A fast software module for online clusterless decoding. *eNeuro*, 12(12). doi:10.1523/ENEURO.0252-24.2025

5. Coulter, M.E., Gillespie, A.K., Chu, J.P., **Denovellis, E.L.**, Nguyen, T.T.K., Liu, D.F., Wadhwani, K., Sharma, B., Wang, K., Deng, X., et al. (2025). Closed-loop modulation of remote hippocampal representations with neurofeedback. *Neuron*, 113(6), 949–961.e3. doi:10.1016/j.neuron.2024.12.023

6. Gillespie, A.K., Astudillo Maya, D.A., **Denovellis, E.L.**, Desse, S., and Frank, L.M. (2024). Neurofeedback training can modulate task-relevant memory replay rate in rats. *eLife*, 12, RP90944. doi:10.7554/eLife.90944.3

7. Joshi, A., **Denovellis, E.L.**, Mankili, A., Meneksedag, Y., Davidson, T.J., Gillespie, A.K., Guidera, J.A., Roumis, D.K., and Frank, L.M. (2023). Dynamic synchronization between hippocampal representations and stepping. *Nature*, 617(7959), 125–131. doi:10.1038/s41586-023-05928-6

8. **Denovellis, E.L.**, Myroshnychenko, M., Sarmashghi, M., and Stephen, E.P. (2022). Spectral Connectivity: a python package for computing multitaper spectral estimates and frequency-domain brain connectivity measures on the CPU and GPU. *Journal of Open Source Software*, 7(80), 4840. doi:10.21105/joss.04840

9. Gillespie, A.K., Astudillo Maya, D.A., **Denovellis, E.L.**, Liu, D.F., Kastner, D.B., Coulter, M.E., Roumis, D.K., Eden, U.T., and Frank, L.M. (2021). Hippocampal replay reflects specific past experiences rather than a plan for subsequent choice. *Neuron*, 109(19), 3149–3163.e6. doi:10.1016/j.neuron.2021.07.029

10. **Denovellis, E.L.**, Gillespie, A.K., Coulter, M.E., Sosa, M., Chung, J.E., Eden, U.T., and Frank, L.M. (2021). Hippocampal replay of experience at real-world speeds. *eLife*, 10, e64505. doi:10.7554/eLife.64505

11. Buschman, T.J.*, **Denovellis, E.L.***, Diogo, C.*, Bullock, D., and Miller, E.K. (2012). Synchronous Oscillatory Neural Ensembles for Rules in the Prefrontal Cortex. *Neuron*, 76(4), 838–846. doi:10.1016/j.neuron.2012.09.029

Preprints

1. Gu, S., Liu, C., Gillespie, A.K., Nevers, R., **Denovellis, E.L.**, Coulter, M.E., Kay, K., and Frank, L.M. (2026). Naturalistic behavior and self-generated neural activity predictive of self-correction. bioRxiv. doi:10.64898/2026.05.26.727951

2. Sun, X., Comrie, A.E., Kahn, A.E., Monroe, E.J., Washington, C.B., Joshi, A., Guidera, J.A., **Denovellis, E.L.**, Krausz, T.A., Zhou, J., et al. (2026). Meta-learning is expressed through altered prefrontal cortical dynamics. bioRxiv. doi:10.64898/2026.01.01.697272

Manuscripts in Preparation

1. Zeng, S., Comrie, A.E., Frank, L.M., Eden, U.T., and **Denovellis, E.L.** (2026). Local goodness-of-fit measures for neural decoding. In preparation

Conference Papers

1. **Denovellis, E.L.**, Frank, L.M., and Eden, U.T. (2019). Characterizing hippocampal replay using hybrid point process state space models. *2019 53rd Asilomar Conference on Signals, Systems, and Computers*, 245–249. doi:10.1109/ieeeconf44664.2019.9048688

Funded Proposals

NIH/NIMH RF1MH130623 (PIs: L. M. Frank, U. T. Eden, A. D. Kaplan) **2022 – 2026**
Statistical machine learning tools for understanding neural ensemble representations and dynamics. BRAIN Initiative: Integration and Analysis of BRAIN Initiative Data; $1.86M. Helped write the Specific Aims and Significance sections; the proposal extends my clusterless decoding methods.

Simons Foundation, SCGB Undergraduate Research Fellowship (SURF) **2020**
Co-wrote the funded proposal as a member of the originating team. Now the Shenoy Undergraduate Research Fellowship in Neuroscience (SURFiN), with 93 fellows in 2026–2027.

Software Packages

Spyglass — Data analysis framework for reproducible neuroscience research. Integrates NWB data standards, DataJoint pipelines, SpikeInterface, and DeepLabCut into a unified platform for multi-lab collaboration. *Co-lead developer.* **121 stars, 62 forks.** Published in *eLife*: doi:10.7554/eLife.108089.3.

spectral_connectivity — Multitaper spectral estimation and frequency-domain connectivity analysis (coherence, Granger causality, phase-lag index) for electrophysiological data. Supports GPU acceleration. *Lead developer.* **137 stars, 49 forks.** Published in *Journal of Open Source Software*: doi:10.21105/joss.04840.

replay_trajectory_classification — State-space models for decoding hippocampal replay trajectories and classifying their dynamics using sorted or clusterless spikes. *Lead developer.* **53 stars, 20 forks.** Published in *eLife*: doi:10.7554/eLife.64505.

non_local_detector — Bayesian state-space models in JAX for decoding non-local neural activity (e.g., hippocampal replay) from sorted or clusterless spikes. Includes continuous/fragmented and multi-environment classifiers. Supports GPU acceleration. *Lead developer.*

ripple_detection — Detection of hippocampal sharp wave ripples and high synchrony events from LFP and multiunit activity, implementing nine published and lab-standard detectors. *Lead developer.* **47 stars, 16 forks.**

track_linearization — Hidden Markov model for mapping 2D animal position onto 1D track graphs (mazes, figure-8s, T-mazes). *Lead developer.*

SpectraVis — Interactive D3.js web application for exploring task-related functional brain connectivity over time and frequency. *Lead developer.* **37 stars, 12 forks.**

RasterVis — Interactive D3.js web application for visualizing spike rasters across neurons, trials, and experimental conditions. *Lead developer.* **12 stars, 4 forks.** Demo: neurophysvis.github.io/RasterVis.

Talks

Invited Talks

NeuroAI for the Hippocampus workshop (session: NeuroAI tools for hippocampal data). Spring Hippocampal Symposium, Verona, Italy, May 2027 (upcoming).

Neuroscience Frontiers, Columbia University, New York, NY, November 2026 (upcoming).

Understanding cognitive representations and dynamics through statistical machine learning. National Centre for Biological Sciences, Bangalore, India, March 2026.

Understanding cognitive representations and dynamics through statistical machine learning. Sainsbury Wellcome Centre/Gatsby Computational Neuroscience Unit, University College London, London, United Kingdom, August 2024.

Understanding cognitive representations and dynamics through statistical machine learning. Computational Neuroscience Center, University of Washington, Seattle, WA, May 2024.

Building Accessible Computational Tools for Modern Systems Neuroscience. Databinge, NeuroImaging and NeuroComputation Centre, University of British Columbia, Vancouver, BC, Canada, February 2024.

Characterizing hippocampal replay using switching point process state space models. Stanford Computational Neuroscience Journal Club, Stanford, CA, August 2020.

Contributed Talks

Characterizing hippocampal replay using hybrid point process filters. IEEE Asilomar Conference on Signals, Systems, and Computers, Pacific Grove, CA, November 2019.

Using state space models to identify latent temporal dynamics of population spiking activity. The 33rd New England Statistics Symposium, Hartford, CT, May 2019.

Better Science Code. MIT Center for Brains, Minds, and Machines, Cambridge, MA, May 2017.

Data visualization tools: from classroom to science. Inter Science of Learning Centers Conference Workshop, San Diego, CA, June 2015.

Synchronous Neural Ensembles for Rules in the Prefrontal Cortex. The 5th Annual Dana and Betty Fisher Retreat of the Picower Institute, South Yarmouth, MA, June 2012.

Neural Dynamics of Cognitive Flexibility. Fall 2011 Picower Plastic Lunch Series, Massachusetts Institute of Technology, Cambridge, MA, October 2011.

Teaching

National Centre for Biological Sciences, Bangalore, India

Visiting Instructor **February – March 2026**

Designed and taught a four-week computational neuroscience course for about 10 graduate students (two 1.5-hour lectures per week), with Jupyter notebook assignments and a final project. Topics: open data use and visualization, spectral analysis of local field potentials, spike-stimulus analysis, neural decoding, and clusterless decoding.

University of California, San Francisco

Guest Lecturer, NS219 Encoder-Decoder Models for Neuroscience **May 2024**

Switching state space models to characterize non-local activity in the hippocampus.

Boston University

Instructor, CELEST Summer Program for Undergraduate Interns **June – August 2015**
Taught Matlab programming course to underrepresented undergraduate students in STEM.

Teaching Fellow, CAS NE 340 **January – May 2014**
Introduction to Computational Models of Skilled Decision and Action. Delivered lectures on task switching and Matlab programming; assisted with computer lab sessions.

Teaching Fellow, CAS CN 570 **January – May 2011**
Neural and Computational Models of Conditioning, Reinforcement, Motivation and Rhythm. Designed final project assignment using a neural network model for visually-guided reaching that integrates motor and visual sensory feedback.

University of California, Santa Barbara

Mathematics Tutor, Campus Learning Assistance Services
September 2005 – June 2007
Tutored students in calculus and linear algebra.

Mentoring & Supervision

University of California, San Francisco

Chris Brozdowski, Computational Scientist **2023 – Present**
Spyglass maintainer: core infrastructure, data export, testing; co-author, Spyglass (2026).

Samuel Bray, Computational Scientist **2023 – Present**
Spyglass development (MoSeq pipeline, DANDI export, NWB ingestion); machine learning for unclustered spike trains (C3PO); co-author, Spyglass (2026).

Boston University

Mehrad Sarmashghi, Ph.D. Student, Division of Systems Engineering **2016 – 2019**
Co-author, Spectral Connectivity (2022).

Training

Inclusive Research Mentor/Manager Training, UCSF **Fall 2023**

Service & Outreach

Reviewing
Ad hoc Reviewer **January 2023 – Present**
Journals: Science, eLife, PLOS Computational Biology, Neural Computation, Journal of Neurophysiology, Journal of Open Source Software. Conferences: COSYNE, IEEE VIS.

Outreach
Originating team, Simons SCGB Undergraduate Research Fellowship (SURF) **June 2020**
Helped propose the program and its format: paid, mentored research in SCGB labs for undergraduates underrepresented in neuroscience. Proposal listed under Funded Proposals.

Co-organizer, CELEST Summer Undergraduate Training Program
May 2013 – June 2015
Organized professional development seminars and led programming tutorials and laboratory tours for undergraduates from underrepresented groups.

Institutional Service
Co-President, CELEST Student Organization **May 2011 – June 2016**
Organized student events and the speaker series; represented students at faculty meetings.

Professional Memberships
Society for Neuroscience **May 2011 – Present**

Conference Abstracts

Coulter, M.E., Gillespie, A.K., Chu, J.P., **Denovellis, E.L.**, Liu, D.F., Nguyen, T.T.K., Wadhwani, K., Sharma, B., Deng, X., Eden, U.T., et al. (2025). Hippocampal neurofeedback promotes generation of spatial representations that jump directly to distant locations. Society for Neuroscience Annual Meeting, San Diego, CA. Program No. NANO011.05

Sun, X., Comrie, A.E., Monroe, E.J., Kahn, A.E., Joshi, A., Guidera, J.A., Tong, L., **Denovellis, E.L.**, Krausz, T.A., Berke, J.D., et al. (2025). Flexible decision-making engages generalizable spiraling dynamical motifs in prefrontal cortex. Computational and Systems Neuroscience (COSYNE), Montreal, QC, Canada. Talk T-21

Comrie, A.E., Monroe, E.J., Kahn, A.E., **Denovellis, E.L.**, Joshi, A., Guidera, J.A., Krausz, T.A., Berke, J.D., Daw, N.D., and Frank, L.M. (2025). Representations of alternative possibilities are flexibly generated to meet cognitive demands. Computational and Systems Neuroscience (COSYNE), Montreal, QC, Canada. Poster 1-076

Coulter, M.E., Gillespie, A.K., Chu, J.P., **Denovellis, E.L.**, Sharma, B., Deng, X., Eden, U.T., Kemere, C.T., and Frank, L.M. (2024). Hippocampal cellular assembly activity during remote representation neurofeedback. Society for Neuroscience Annual Meeting, Chicago, IL. Program No. PSTR045.17

Sun, X., Comrie, A.E., Monroe, E.J., Kahn, A.E., Joshi, A., Guidera, J.A., Tong, L., **Denovellis, E.L.**, Krausz, T.A., Shin, D., et al. (2024). Dynamic value codes in the medial prefrontal cortex to inform decision-making. Computational and Systems Neuroscience (COSYNE), Lisbon, Portugal

Coulter, M.E., Gillespie, A.K., Chu, J.P., **Denovellis, E.L.**, Nguyen, T.T.K., Liu, D.F., Wadhwani, K., Sharma, B., Wang, K., Deng, X., et al. (2023). Closed-loop modulation of hippocampal representations through neurofeedback. Society for Neuroscience Annual Meeting, Washington, DC. Online

Comrie, A.E., Monroe, E.J., **Denovellis, E.L.**, Joshi, A., Guidera, J.A., Kahn, A.E., Krausz, T.A., Berke, J.D., Daw, N.D., and Frank, L.M. (2023). Dynamic engagement of non-local spatial representations in the hippocampus during value-guided foraging decisions. Society for Neuroscience Annual Meeting, Washington, DC. Online

Krausz, T.A., Egert, D., Comrie, A.E., **Denovellis, E.L.**, Daw, N.D., Frank, L.M., and Berke, J.D. (2023). Hippocampal non-local representations retrieve and update dopaminergic place values. Society for Neuroscience Annual Meeting, Washington, DC. Online

Krausz, T.A., Egert, D., Comrie, A.E., **Denovellis, E.L.**, Daw, N.D., Frank, L.M., and Berke, J.D. (2023). Hippocampal non-local representations retrieve and update dopaminergic place values. Gordon Research Conference: Catecholamines, Castelldefels, Spain

Guidera, J.A., Gramling, D.P., **Denovellis, E.L.**, Comrie, A.E., Joshi, A., Zhou, J., Thompson, P., Hernandez, J., Yorita, A., Haque, R., et al. (2023). A region-specific code for specific and generalized representations across prefrontal cortex and hippocampus. Computational and Systems Neuroscience (COSYNE), Montreal, QC, Canada

Denovellis, E.L., Guidera, J.A., and Frank, L.M. (2023). Switching state-space models enable decoding of replay across multiple spatial environments. Computational and Systems Neuroscience (COSYNE), Montreal, QC, Canada

Denovellis, E.L., Joshi, A., Eden, U.T., and Frank, L.M. (2022). Hippocampal replay outside of sharp wave ripples. Society for Neuroscience Annual Meeting, San Diego, CA. Program No. 742.04. Online

Chu, J.P., Coulter, M.E., Liu, D.F., **Denovellis, E.L.**, Frank, L.M., and Kemere, C.T. (2022). Real-time decoding with state space models of neural activity. Society for Neuroscience Annual Meeting, San Diego, CA. Program No. 742.08. Online

Gillespie, A.K., Astudillo Maya, D.A., **Denovellis, E.L.**, Desse, S., Eden, U.T., and Frank, L.M. (2022). Real-time feedback can promote task-relevant memory replay. Society for Neuroscience Annual Meeting, San Diego, CA. Program No. 742.05. Online

Lee, K.H., **Denovellis, E.L.**, Ly, R., Magland, J., Soules, J., Comrie, A.E., Guidera, J.A., Nevers, R., Gramling, D.P., Adenekan, P., et al. (2022). Spyglass: a data analysis framework for reproducible and shareable neuroscience research. Society for Neuroscience Annual Meeting, San Diego, CA. Program No. 742.16. Online

Denovellis, E.L., and Frank, L.M. (2022). Confidently decoding multiple spatial environments in hippocampal replay. AREADNE: Research in Encoding and Decoding of Neural Ensembles, Santorini, Greece. Program No. 66. Travel award. Online

Denovellis, E.L., Gillespie, A.K., Coulter, M.E., Sosa, M., Eden, U.T., and Frank, L.M. (2021). Hippocampal replay of experience at real-world speeds. Society for Neuroscience Annual Meeting, Chicago, IL. Program No. 842.02. Online

Gillespie, A.K., Astudillo Maya, D.A., **Denovellis, E.L.**, Liu, D.F., Kastner, D.B., Coulter, M.E., Roumis, D.K., Eden, U.T., and Frank, L.M. (2021). Hippocampal replay reflects specific past experiences rather than a plan for subsequent choice. Society for Neuroscience Annual Meeting, Chicago, IL. Program No. 842.01. Online

Denovellis, E.L., Gillespie, A.K., Coulter, M.E., Frank, L.M., and Eden, U.T. (2019). A state space model for characterizing replay dynamics. Society for Neuroscience Annual Meeting, Chicago, IL. Program No. 162.04. Online

Gillespie, A.K., Astudillo Maya, D.A., Liu, D.F., Coulter, M.E., **Denovellis, E.L.**, Desse, S., Roumis, D.K., Eden, U.T., and Frank, L.M. (2019). Operant conditioning of hippocampal sharp wave ripples. Society for Neuroscience Annual Meeting, Chicago, IL. Program No. 162.02. Online

Denovellis, E.L., Gillespie, A.K., Coulter, M.E., Frank, L.M., and Eden, U.T. (2019). A state space model for characterizing trajectory dynamics of non-local spatial firing in hippocampus. Ninth International Workshop on Statistical Analysis of Neural Data (SAND9), Pittsburgh, PA

Denovellis, E.L., Stephen, E.P., Eden, U.T., and Kramer, M.A. (2018). Interactive data visualization for electrophysiological data. Society for Neuroscience Annual Meeting, San Diego, CA. Program No. 703.20. Online

Buschman, T.J., **Denovellis, E.L.**, Diogo, C., Bullock, D., and Miller, E.K. (2012). Dynamic, synchronous, sub-networks in prefrontal cortex encode stimulus-response rules. Society for Neuroscience Annual Meeting, New Orleans, LA. Program No. 599.10. Online

Denovellis, E.L., Buschman, T.J., Diogo, C., Bullock, D., and Miller, E.K. (2012). Point process models of anterior cingulate and dorsolateral prefrontal cortical neurons during cognitive control. Society for Neuroscience Annual Meeting, New Orleans, LA. Program No. 599.12. Online

Buschman, T.J., **Denovellis, E.L.**, Diogo, C., Bullock, D., and Miller, E.K. (2012). Dynamic networks in frontal cortex support the cognitive flexibility to switch between rules. Computational and Systems Neuroscience (COSYNE), Salt Lake City, UT

Denovellis, E.L., Buschman, T.J., Diogo, C., Bullock, D., and Miller, E.K. (2011). Rule-based task switching in the anterior cingulate and prefrontal cortex. Society for Neuroscience Annual Meeting, Washington, DC. Program No. 405.18. Online

Other Experience

Pension Consultant I, Mercer Advisors **October 2007 – July 2009**